The Journal of Open Source Software

Scikit-Longitudinal: A Machine Learning Library for Longitudinal Classification in Python

Simon Provost [1] **and Alex A. Freitas** [1]

1 School of Computing, University of Kent, Canterbury, United Kingdom

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 09 May 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Introduction

Longitudinal data, characterised by repeated measurements of variables over time, presents unique challenges and opportunities in machine learning. This paper introduces `Scikit-Longitudinal`, a Python library designed to address these challenges by providing a comprehensive set of tools for longitudinal data classification. Built to integrate flawlessly with the widely-used `Scikit-Learn` library, `Scikit-Longitudinal` offers a robust solution for researchers and practitioners working with longitudinal datasets.

Summary

`Scikit-Longitudinal` is an open-source Python library that enhances machine learning for longitudinal data classification and integrates flawlessly with the `Scikit-Learn` environment (Pedregosa et al., 2011).

Longitudinal data, which consists of repeated measurements of variables across time points (referred to as waves (Ribeiro & Freitas, 2019)), is extensively utilised in fields such as medicine, social sciences, and economics. Unlike standard tabular datasets, it contains temporal relationships that necessitate specialised processing (Kelloway & Francis, 2012).

`Scikit-Longitudinal` addresses this with a novel library that includes:

- **Data Preparation**: Utilities such as `LongitudinalDataset` for loading and structuring data, defining temporal feature groups, and other techniques.
- **Data Transformation**: Methods to treat the temporal aspect of tabular data, by either (1) flattening the data into a static representation (i.e. ignoring time indices) for standard machine learning to be performed (e.g., `MarWavTimeMinus`, or `SepWav`), or (2) keeping the temporal structure (e.g., `MerWavTimePlus`), yet saving it for later use in longitudinal-data-aware `preprocessing` or `estimators` steps (Ribeiro & Freitas, 2019).
- **Preprocessing**: Longitudinal-data-aware feature selection primitives, such as `CFS-Per-Group` (Pomsuwan & Freitas, 2017), utilising the temporal information in the data to proceed with feature selection techniques (see feature selection review in (Theng & Bhoyar, 2024)).
- **Estimators**: Longitudinal-data-aware classifiers (Kotsiantis et al., 2007) (Ribeiro & Freitas, 2019), such as `LexicoRandomForestClassifier` (Ribeiro & Freitas, 2024), `LexicoGradientBoostingClassifier`, and `NestedTreesClassifier` (Ovchinnik et al., 2022), which leverage the temporal structure of the data to ideally enhance classification performance.

In total, the library implements 1 data preparation method, 4 data transformation methods, 1 preprocessing method, and 6 estimators, 2 of which have been published as stand

alone methods in the literature (the above named `LexicoRandomForestClassifier` and `NestedTreesClassifier` methods).

`Scikit-Longitudinal` emphasises highly-typed, Pythonic code, with substantial test coverage (over 88%) and comprehensive documentation (over 72%), ensuring reliability and extensibility for researchers and practitioners in longitudinal data analysis.

Finally, `Scikit-Longitudinal` can also be abbreviated `Sklong`, and is available on PyPI. Feel free to explore the official documentation for various installation methods, including locally, via `Colab`, `Marimo`, and others.

Longitudinal Classification

Longitudinal classification is a variant of the standard classification task where the data includes features taking values at multiple time points / "waves" (Ribeiro & Freitas, 2019), e.g., cholesterol values measured at different waves. Longitudinal classification is particularly relevant in biomedical applications, since biomedical data about patients is often collected across long time periods.

The challenge is to learn a model that predicts the class label (Y) for an instance while accounting for the evolution of features' values over time, i.e., to learn a predictive model (classifier function) of the form:

$$Y \leftarrow f(X_{1,1}, X_{1,2}, \dots, X_{1,T}, \dots, X_{K,1}, X_{K,2}, \dots, X_{K,T})$$

where $X_{i,j}$, for $i = 1, \dots, K$ and $j = 1, \dots, T$, is the value of the i-th feature at the j-th wave (time point), K is the number of features, and T is the number of waves. The classifier function $f(\cdot)$ can either operate on a transformed version of the data where temporal information is "flattened", allowing the application of standard machine learning algorithms, or handle the temporal dependencies between a feature's values across time and between different features directly. Note that, in the type of longitudinal classification task for which `Sklong` was designed, the features are longitudinal, but the class variable is not; i.e., the goal is to predict the class label of an instance at a single time point (usually the most recent wave).

Approaches to Longitudinal Classification

There are two broad approaches for coping with longitudinal data (Ribeiro & Freitas, 2019):

1. **Data Transformation**
   This approach involves preprocessing methods that convert longitudinal data into a standard, "flattened" non-longitudinal format. This transformation enables the use of any standard, non-longitudinal classification algorithm on the data. However, this approach may result in the loss of relevant information regarding how a feature's values change over time.

2. **Algorithm Adaptation**
   This approach entails modifying classification algorithms to directly handle temporal dependencies inherent in longitudinal datasets. This preserves the temporal dynamics of the data but may require more complex modelling techniques.

Statement of Need

The temporal nature of longitudinal data poses significant challenges for conventional machine learning techniques, which typically assume independence among observations. Standard libraries such as `Scikit-learn` do not inherently support longitudinal data, resulting in inefficient or inaccurate analyses. Consequently, there is a pressing need for a dedicated library that

DRAFT

offers a comprehensive suite of tools for longitudinal data, encompassing data preparation, transformation, preprocessing, and estimation (model learning).

The R programming language (compared to the Python sphere) has seen the development of numerous statistical packages for longitudinal data over the years, including `nlme` (Pinheiro & Bates, 2000) and `GPBoost` (Sigrist, 2022). However, these packages are primarily tailored for statistical models like linear mixed-effects models and do not seamlessly integrate with machine learning workflows. Furthermore, they often lack the flexibility and modularity characteristic of `Scikit-learn`, complicating their incorporation into existing machine learning pipelines (which are predominantly developed in Python). This underscores the need for a library (primarily in Python) that not only provides tools for longitudinal data but also ensures compatibility with established machine learning frameworks.

Additionally, although there are a handful of machine learning-centric tools like Auto-Prognosis (Alaa & Schaar, 2018) that to some extent address longitudinal classification, they often do not match the simplicity and out-of-the-box usability of `Scikit-learn`. This gap motivated the development of `Scikit-Longitudinal`, which aims to provide a straightforward and integrated solution for longitudinal classification within the `Scikit-learn` ecosystem.

As a result, the introduction of `Scikit-longitudinal` creates an ecosystem for the longitudinal data analysis community, providing longitudinal-data-processing primitives that can be used for implementing existing and new longitudinal classification algorithms or pipelines, in a fast and reliable way (from a software development perspective), whilst also providing compatibility with the well-known `Scikit-Learn` library.

The story behind `Scikit-Longitudinal`

`Scikit-Longitudinal` (`Sklong`) was initially developed as part of a PhD project at the University of Kent, United Kingdom. Specifically, it was created as a library to be used as part of an Automated Machine Learning (AutoML) system for longitudinal classification, called `Auto-Sklong` (Provost & Freitas, 2024).

`Auto-Sklong` is among the first AutoML systems designed for longitudinal classification. It offers a user-friendly interface, and it leverages the functionality of both `Sklong` and `Scikit-learn` for automating the choice of the best longitudinal classification pipeline (and its best hyperparameter settings) for a given input longitudinal classification dataset.

Although `Auto-Sklong` successfully utilises both the `Sklong` and the `Scikit-learn` libraries, the `Sklong` library can also be used stand alone as an out-of-the-box software tool for longitudinal data analysis.

Limitations and Future Work

At present, `Scikit-Longitudinal` primarily focuses on the classification task and does not yet include support for regression or neural network models. Future development could expand the library to encompass these areas, thereby broadening its applicability.

Acknowledgements

We extend our gratitude to the authors and contributors of `Scikit-learn` for their pioneering work in the machine learning community. We also acknowledge the NeuroData team for their contributions to `Scikit-Tree` (Li et al., 2024), which facilitates the customisation of `Scikit-learn`'s Cython trees. Furthermore, we are indebted to the researchers whose work on longitudinal machine learning primitives has been instrumental in the development of `Scikit-Longitudinal`, including Dr. Tossapol Pomsuwan (Pomsuwan, 2018; Pomsuwan &

Freitas, 2017), Dr. Sergey Ovchinnik and Dr. Fernando Otero (Ovchinnik et al., 2022), and Dr. Caio Ribeiro (Ribeiro, 2022; Ribeiro & Freitas, 2019, 2024).

References

Alaa, A., & Schaar, M. (2018). Autoprognosis: Automated clinical prognostic modeling via bayesian optimization with structured kernel learning. *Proceedings of the International Conference on Machine Learning*, 139–148.

Kelloway, E. K., & Francis, L. (2012). Longitudinal research and data analysis. In *Research methods in occupational health psychology* (pp. 374–394). Routledge.

Kotsiantis, S. B., Zaharakis, I., Pintelas, P., & others. (2007). Supervised machine learning: A review of classification techniques. *Emerging Artificial Intelligence Applications in Computer Engineering*, *160*(1), 3–24.

Li, A., Panda, S., Xu, H., & Ogihara, I. (2024). *treeple: Modern decision-trees compatible with scikit-learn in Python.*

Ovchinnik, S., Otero, F., & Freitas, A. A. (2022). Nested trees for longitudinal classification. *Proceedings of the 37th ACM/SIGAPP Symposium on Applied Computing*, 441–444.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., & others. (2011). Scikit-learn: Machine learning in python. *The Journal of Machine Learning Research*, *12*, 2825–2830.

Pinheiro, J., & Bates, D. (2000). *Mixed-effects models in s and s-PLUS*. Springer science & business media.

Pomsuwan, T. (2018). *Feature selection for the classification of longitudinal human ageing data* [Master by Research thesis]. School of Computing, University of Kent.

Pomsuwan, T., & Freitas, A. A. (2017). Feature selection for the classification of longitudinal human ageing data. *Proceedings of the 2017 IEEE International Conference on Data Mining Workshops (ICDMW)*, 739–746.

Provost, S., & Freitas, A. A. (2024). Auto-sklong: A new AutoML system for longitudinal classification. *Proceedings of the 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 3645–3650.

Ribeiro, C. (2022). *New longitudinal classification approaches and applications to age-related disease data* [Ph.D. Thesis]. School of Computing, University of Kent.

Ribeiro, C., & Freitas, A. A. (2019). A mini-survey of supervised machine learning approaches for coping with ageing-related longitudinal datasets. *Proceedings of the 3rd Workshop on AI for Aging, Rehabilitation and Independent Assisted Living (ARIAL), Held as Part of IJCAI-2019*.

Ribeiro, C., & Freitas, A. A. (2024). A lexicographic optimisation approach to promote more recent features on longitudinal decision-tree-based classifiers: Applications to the english longitudinal study of ageing. *Artificial Intelligence Review*, *57*, Article 84.

Sigrist, F. (2022). Gaussian process boosting. *Journal of Machine Learning Research*, *23*(232), 1–46.

Theng, D., & Bhoyar, K. K. (2024). Feature selection techniques for machine learning: A survey of more than two decades of research. *Knowledge and Information Systems*, *66*(3), 1575–1637.

DRAFT